

Question	Answer	Marks
7(a)	(magnetic) force (always) normal to velocity/direction of motion	M1
	(magnitude of magnetic) force constant or speed is constant/kinetic energy is constant	M1
	so provides the centripetal force	A1
7(b)	increase in KE = loss in PE or $\frac{1}{2}mv^2 = qV$	M1
	$p = mv$ with algebra leading to $p = \sqrt{(2mqV)}$	A1
7(c)	$Bqv = mv^2 / r$ $mv = Bqr$ or $p = Bqr$	C1
	$(2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 120)^{1/2} = B \times 1.60 \times 10^{-19} \times 0.074$	C1
	$B = 5.0 \times 10^{-4} \text{ T}$	A1
7(d)	greater momentum	M1
	($p = Bqr$ and) so r increased	A1

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8	strong (uniform) magnetic field	B1
	* <u>nuclei</u> precess/rotate about field (direction)	
	radio frequency pulse/RF pulse (applied)	B1
	* RF or pulse is at Larmor frequency / frequency of precession	
	causes resonance / excitation (of nuclei)/nuclei to absorb energy	B1
	on relaxation/de-excitation, nuclei emit RF/pulse	B1
	* (emitted) RF/pulse detected and processed	
	non-uniform field (superposed on uniform field)	B1
	allows positions of (resonating) <u>nuclei</u> to be determined	B1
	* allows for position of detection to be changed/different slices to be studied	
	<i>max. 2 of additional detail points marked *</i>	B2